MEDIUM · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Mixed

01

10 answers · Rationale-rich · Teach from doc

Q1

D

D. 162

Choice D is correct. It's given that triangle XYZ is similar to triangle TUV. Therefore, each side of triangle XYZ is k times its corresponding side of triangle TUV, where k is a constant. It follows that the perimeter of triangle XYZ is k times the perimeter of triangle TUV. It's also given that $\overline{TU}$ corresponds to $\overline{XY}$ and the length of $\overline{TU}$ is 18. Let x represent the length of $\overline{XY}$. It follows that $x = 18k$. The table shows that the perimeters of triangles TUV and XYZ are 37 and 333, respectively. It follows that $333 = 37k$, or $9 = k$. Substituting 9 for k in the equation $x = 18k$ yields $x = 189$, or $x = 162$. Therefore, the length of $\overline{XY}$ is 162.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{TU}$, not the length of $\overline{XY}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q2

4205

The correct answer is 4,205. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of 5 congruent square faces. It's also given that the length of each edge is 29 inches. Let s represent the length of an edge of a square. It follows that the area of a square is equal to s^2. Therefore, the area of each of the 5 square faces is equal to 29^2, or 841 , square inches. Since the box consists of 5 congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is 5841, or 4,205, square inches.

Q3

D

D. $930 < w < 1,295$

Choice D is correct. It's given that the elephant weighs 200 pounds at birth and gains more than 2 pounds but less than 3 pounds per day during its first year. The inequality $200 + 2d < w < 200 + 3d$ represents this situation, where d is the number of days after birth. Substituting 365 for d in the inequality gives $200 + 2(365) < w < 200 + 3(365)$, or $930 < w < 1,295$.

Choice A is incorrect and may result from solving the inequality $200(2) < w < 200(3)$. Choice B is incorrect and may result from solving the inequality for a weight range of more than 1 pound but less than 2 pounds: $200 + 1(365) < w < 200 + 2(365)$. Choice C is incorrect and may result from calculating the possible weight gained by the elephant during the first year without adding the 200 pounds the elephant weighed at birth.

Q4

-2

The correct answer is -2. It's given that a number x is at most 17 less than 5 times the value of y, or $x \leq 5y - 17$. Substituting 3 for y in this inequality yields $x \leq 53 - 17$, or $x \leq -2$. Thus, if the value of y is 3, the greatest possible value of x is -2.

Q5

A

A. Zero

Choice A is correct. A system of two linear equations in two variables, x and y, has zero points of intersection if the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the y-coordinates of their y-intercepts, b, are different and are parallel if their slopes, m, are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1, respectively, and the values of m are both 2. Since the values of b are different, the graphs of these lines have different y-coordinates of the y-intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the xy-plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and the same y-coordinate of the y-intercept. Since the given equations represent lines with different y-coordinates of their y-intercepts, there are not infinitely many intersection points.

Q6

9

The correct answer is 9. The given expression can be rewritten as $5x^3 - 3 + -1 \text{-} 4x^3 + 8$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + -8$, which is equivalent to $5x^3 + 4x^3 + -3 + -8$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $5x^3 - 3 - -4x^3 + 8$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of x^3 in these two expressions must be equivalent, and the value of b must be 9.

Q7

6

The correct answer is 6. It's given that $y = 4x$ and $y = x^2 - 12$. Since $y = 4x$, substituting $4x$ for y in the second equation of the given system yields $4x = x^2 - 12$. Subtracting $4x$ from both sides of this equation yields $0 = x^2 - 4x - 12$. This equation can be rewritten as $0 = x - 6x + 2$. By the zero product property, $x - 6 = 0$ or $x + 2 = 0$. Adding 6 to both sides of the equation $x - 6 = 0$ yields $x = 6$. Subtracting 2 from both sides of the equation $x + 2 = 0$ yields $x = -2$. Therefore, solutions to the given system of equations occur when $x = 6$ and when $x = -2$. It's given that a solution to the given system of equations is x, y, where $x > 0$. Since 6 is greater than 0, it follows that the value of x is 6.

Q8

C

C. $2, 0$

Choice C is correct. The second equation in the given system of equations is $y = 6x - 12$. Substituting $6x - 12$ for y in the first equation of the given system yields $6x - 12 = x - 2x + 4$. Factoring 6 out of the left-hand side of this equation yields $6x - 2 = x - 2x + 4$. An expression with a factor of the form $x - a$ is equal to zero when $x = a$. Each side of this equation has a factor of $x - 2$, so each side of the equation is equal to zero when $x = 2$. Substituting 2 for x into the equation $6x - 2 = x - 2x + 4$ yields $62 - 2 = 2 - 22 + 4$, or $0 = 0$, which is true. Substituting 2 for x into the second equation in the given system of equations yields $y = 62 - 12$, or $y = 0$. Therefore, the solution to the system of equations is the ordered pair $2, 0$.

Choice A is incorrect and may result from switching the order of the solutions for x and y.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

A

A. 3

Choice A is correct. Since the distance is equal to the amount of time multiplied by a constant, the given equation $d = 55t$ represents a proportional relationship between distance and time in this situation. Since $9k = 3 \cdot 3k$, the time when $t = 9k$ hours is 3 times the time when $t = 3k$ hours. Therefore, the distance traveled after $9k$ hours is 3 times the distance after $3k$ hours.

Choices B and D are incorrect and may result from interpreting the proportional relationship between time and distance as additive rather than multiplicative. Choice C is incorrect and may result from an arithmetic error.

Q10

C

C. 2.4

Choice C is correct. At $x = 32$, the line of best fit has a y-value between 2 and 3. The only choice with a value between 2 and 3 is choice C.

Choice A is incorrect. This is the difference between the y-value predicted by the line of best fit and the actual y-value at $x = 32$ rather than the y-value predicted by the line of best fit at $x = 32$.

Choice B is incorrect. This is the y-value predicted by the line of best fit at $x = 31$ rather than at $x = 32$.

Choice D is incorrect. This is the y-value predicted by the line of best fit at $x = 33$ rather than at $x = 32$.

Notes

Accept equivalent forms (e.g. `0.25` ≡ `1/4`). For grid-in items, decimal and fraction answers are both valid.

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